reg vs clf → p1.py

```
import pandas as pd
import matplotlib.pyplot as plt

data = pd.read_csv("age_sub.csv")
print(data)

plt.scatter(data["age"], data["sub"], color="red")
plt.show()
```

soln using LR → p2.py

```
import pandas as pd
from sklearn.linear_model import LogisticRegression

data = pd.read_csv("age_sub.csv")
feature = data[["age"]]
target = data["sub"]

model = LogisticRegression()
model.fit(feature.values, target)

age = float(input("enter age "))
result = model.predict([[age]])
print(result)
```

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Internal working using sigmoid → p3.py

```
import pandas as pd
from sklearn.linear_model import LogisticRegression

data = pd.read_csv("age_sub.csv")
feature = data[["age"]]
target = data["sub"]

model = LogisticRegression()
model.fit(feature.values, target)
print(model.classes_)

b0 = model.intercept_
b1 = model.coef_[0]

age = float(input("enter age "))
result = 1 / ( 1 + 2.71 ** (-1 * (b0 + b1 * age)) )
print(result)

if result >= 0.5:
	print("yes")
else:
	print("no")
```

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 # internal working → using predict_proba → p4.py

```
import pandas as pd
from sklearn.linear_model import LogisticRegression

data = pd.read_csv("age_sub.csv")
feature = data[["age"]]
target = data["sub"]

model = LogisticRegression()
model.fit(feature.values, target)

age = float(input("enter age "))
result = model.predict([[age]])
print(result)

a1 = model.predict_proba([[age]])          # 2 d list
a2 = a1.ravel()                            # 1 d list
print(a2)
pno = round(a2[0] * 100, 2)                # no
pyes = round(a2[1] * 100, 2)               # yes

print("pno = ", pno,  "%", " pyes = ", pyes, "%")
```

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how to calculate → confusion matrix, classification report and accuracy

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix, classification_report

data = pd.read_csv("age_sub.csv")

feature = data[["age"]]
target = data["sub"]

x_train, x_test, y_train, y_test = train_test_split(feature.values, target, test_size=0.4)
print(x_train)
print(y_train)

print(x_test)
print(y_test)

model = LogisticRegression()
model.fit(feature.values, target)

y_pred = model.predict(x_test)
print(y_pred)

input()

cf = confusion_matrix(y_test, y_pred)
print(cf)

input()

cr = classification_report(y_test, y_pred)
print(cr)
```

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